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Mathematical Customer Lifetime Value (CLV) Projection via BG/NBD Model from Scratch

Abstract

This report accompanies a BG/NBD implementation from first principles using NumPy and SciPy's L-BFGS-B optimizer.

Introduction

Customer Lifetime Value (CLV) estimates the long-term value generated by customers. This project implements the BG/NBD model from scratch without specialized marketing libraries.

Theory

The BG/NBD model assumes Poisson purchasing, Gamma-distributed transaction rates, Beta-distributed dropout probabilities, and customer heterogeneity.

Objectives

- Implement BG/NBD from scratch
- Estimate r, alpha, a, b using MLE
- Predict future transactions
- Project CLV

Methodology

1. Prepare transaction summaries (x, t_x, T).
2. Compute vectorized negative log-likelihood.

3. Optimize parameters using SciPy L-BFGS-B.
4. Predict expected transactions and CLV.

Dataset

Synthetic transaction logs are used to validate parameter recovery. The same pipeline can be applied to real transaction datasets after converting them into (x, t_x, T) summaries.

Likelihood

The BG/NBD log-likelihood is maximized by minimizing the negative log-likelihood over all customers using Maximum Likelihood Estimation (MLE).

Complexity Analysis

Vectorized NumPy operations provide O(N) likelihood evaluation per optimizer iteration while avoiding slow Python loops, making the implementation scalable.

Results

The implementation estimates r, alpha, a, and b, computes probability alive, expected future transactions, and projected CLV.

Conclusion

The project demonstrates an efficient, numerically stable BG/NBD implementation suitable for educational and research purposes.

References

1. Fader, Hardie & Lee (2005), Marketing Science.
2. SciPy Documentation.
3. NumPy Documentation.